Prakhathi Murugesan

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SUMMARY

Senior Front-End Software Developer with 6+ years of experience building accessible, scalable, and user-centric web applications. Expert in **React, TypeScript**, and modern design systems with proven track record implementing **WCAG 2.1 A/AA** standards across enterprise applications. Specialized in developing and maintaining **design systems**, **component libraries**, and leveraging **AI tools** to streamline design-to-code workflows. Currently enhancing payroll application at ADP serving thousands of users, with deep expertise in **accessibility**, performance optimization, and cross-functional collaboration in Agile environments.

EDUCATION

University at Buffalo, The State University of New York — Buffalo, NY, USA — **Aug 2019 - Feb 2021**
Master of Science in Computer Science — GPA – 3.78 /4.0

Coimbatore Institute of Technology — Coimbatore, India — **Jul 2012 - May 2016**
Bachelor of Technology in Information Technology — GPA – 3.7/4.0

TECHNICAL SKILLS

Front-End Expertise: React, TypeScript, JavaScript (ES6+), HTML5, CSS3, SCSS/SASS, LESS, Redux, Redux-Saga
Design Systems & Accessibility: Storybook, Style-Dictionary, Design Tokens, WCAG 2.1 A/AA, ARIA, Semantic HTML, Screen Reader Optimization, Keyboard Navigation, Color Contrast Optimization
AI & Automation Tools: Claude AI, AI-assisted design-to-code pipelines, automated component generation
UI Frameworks & Libraries: Material-UI, Bootstrap, React-Native, jQuery
Architecture & Performance: Component Libraries, Reusable UI Frameworks, Build Optimization (Webpack), Code Splitting, Performance Profiling
Visualization & Theming: D3.js, Recharts, Google Charts, Dark/Light Theme Implementation
Testing & Quality: Jest, Enzyme, Jasmine, Test-Driven Development (TDD), Unit Testing
Back-End & APIs: Node.js, Django, REST APIs, JSON, XML
Databases: PostgreSQL, MySQL, MSSQL **Development Tools:** Git, GitHub, SVN, Jira, Trello, Figma

PROFESSIONAL EXPERIENCE

Sr. Application Developer, ADP Inc. — New **Jersey**, USA — **Apr 2024 - Present**
- Leveraged a2ui agent to generate accessible payroll forms compliant with design system and WCAG 2.1 AA standards.

Sr. Application Developer, ADP Inc. — Toronto, Canada — **Mar 2024 - Apr 2024**
- Executed the comprehensive migration and restructuring of the frontend application, successfully transitioning legacy components and architecture to a modernized mono repository structure.
- Developed and refactored complex interactive components including time tracking interfaces, infinite scroll timelines, and dynamic form controls, while establishing **error boundaries** and improving application stability and usability.
- Leveraged **AI tools (Claude)** to accelerate design-to-code workflows, reducing component development time by 35% while maintaining design system consistency

Sr. Application Developer, ADP Inc. — New Jersey, USA — **Sept 2022 - Mar 2024**
- Architected and developed **progressive, embeddable forms** using React and TypeScript for enterprise clients, enabling seamless integration into third-party applications serving several end users
- Supported **dark/light theme** mode of the application by creating design system using style-dictionary.
- Built a **single-page application (SPA)** to upload files which was a critical part of Ireland's launch of the application.
- Reviewed **build tool** choices and replaced heavier packages with lightweight, efficient ones that are 20x faster in compilation.

Application Developer, ADP Inc. — New Jersey, USA — **Mar 2021 - Aug 2022**
[React JS, Redux, Typescript, Storybook, a11y (accessibility)]
- Enhanced accessibility of the application by providing screen reader support using **semantic HTML, ARIA, SR labels** and Keyboard navigation; building fluid layout to support dynamic font size and optimizing color contrast.
- Resolved crucial product bugs by effectively analyzing existing code and upgraded flawed features into user-friendly features.

Software Engineer Intern, Twine Labs — Remote, USA — **Jun 2020 - Dec 2020**
[React JS, Redux, Django, PostgreSQL, AWS, D3.js]
- Built a new Analytics dashboard and re-vamped the product's forum segment to boost user engagement by **30%.**
- Developed interactive user-facing features with in-built carousel and animations by translating **Figma wireframes.**
- Enhanced the performance of the product by 10% by reimplementing redux-based functionalities into custom **React hooks**.
- Executed **Django models** to notify the team in Slack about prospects visiting the website and helped in reducing TAT.

Software Engineer, Tavant Technologies — Bengaluru, India — **Aug 2016 - Jul 2019**
- Developed a web service that automated the client's monitoring system of their customers' mortgages and repayments.
- Revamped and optimized the client's existing website into a cross-browser and responsive website to increase efficiency.
- Spearheaded an internal project that built an application to support HR activities and monitor employees' performance.
- Incorporated module bundlers such as **webpack** into the build environment to reduce the page load time by 40%.
- Adopted **Test-Driven Development** (TDD) to deliver high-quality codes with 100% unit test coverage.
- Evaluated the user stories and allocated tasks to the team based on effort and time estimates following **Agile methodology**.
- Collaborated with other teams' stakeholders to discuss the requirements, feasibility and to resolve operational challenges.